Heat Conduction Simulation and Physics Informed Neural Networks (PINN)

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Abstract

This documentation presents numerical studies for the 2D heat conduction equation. The heat equation is solved using classical numerical methods, such as explicit Euler and implicit Crank-Nicolson, as well as Physics Informed Neural Networks (PINNs). Comparisons between these methods are provided, highlighting stability, accuracy, and computational efficiency.

Contents

1 The classical Heat Equation

1.1 The Energy Balance and the Derivation of the Heat Equation

We now derive the heat equation from the energy balance. For further reading see [5, pp 207] and [8, pp 105] Heat conduction describes the transfer of thermal energy in materials and is governed by the heat equation. The mechanism of heat conduction takes places on the scale of atoms and molecules and their local motion as mechanism to transfer heat. We consider an open spatial domain $\Omega \subset \mathbb{R}^d$ where the heat conduction is investigated. Let $R \subset \Omega$ be an arbitrary subdomain of Ω with the following assumptions:

1. **Regularity:** R is regular, so that all following integrals and operations on R are well-defined.
2. **Absence of particle motion:** R does not move over time, i.e. macroscopic motion of particles inside R is not allowed.

The heat equation can be derived from energy balance as follows. The change in time of the internal energy $\mathscr{E}(R)$ is given by the heat flux crossing the boundary of R and the heat sources and sinks in R, i.e.

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathscr{E}(R) = \frac{\mathrm{d}}{\mathrm{d}t}\int_R \varrho u \mathrm{d}V = \int_R \varrho f \mathrm{d}V - \int_{\partial R} \boldsymbol{q} \cdot \boldsymbol{\nu}_{\partial R} \mathrm{d}A.$$

The used symbols have the following meaning:

Symbol	SI Unit	Description
ϱ	$\frac{\mathrm{kg}}{\mathrm{m}^d}$	Mass density
u	$\frac{\mathrm{J}}{\mathrm{kg}}$	Specific internal energy
f	$\frac{\mathrm{W}}{\mathrm{kg}}$	Specific hat sources
$\boldsymbol{q}$	$\frac{\mathrm{W}}{\mathrm{m}^{d-1}}$	Heat flux per area
$\boldsymbol{\nu}_{\partial R}$	1	Outer unit normal at the boundary of R

Table 1: Physical quantities.

Since R is regular we can interchange time derivation and integration and apply the Gauss-Green theorem on the boundary integral to obtain

$$\int_R \frac{\partial}{\partial t}(\varrho u)\,\mathrm{d}V = \int_R (\varrho f - \mathrm{div}(\boldsymbol{q}))\,\mathrm{d}V, \text{ or}$$

$$\frac{1}{\mathrm{meas}(R)}\int_R \frac{\partial}{\partial t}(\varrho u)\,\mathrm{d}V = \frac{1}{\mathrm{meas}(R)}\int_R (\varrho f - \mathrm{div}(\boldsymbol{q}))\,\mathrm{d}V.$$

Since R is an arbitrary sample volume, we can drop the integration to get

$$\frac{\partial}{\partial t}(\varrho u) = \varrho f - \mathrm{div}(\boldsymbol{q}).$$

We now fix the following assumptions:

(A1) Heat conduction is the only mechanism of heat transfer. That means convection and radiation are not present.

(A2) Heat sources and sinks in the bulk are nor present.

(A3) The specific internal energy is given by $u = c_P \cdot T$, where c_P is the specific heat capacity at constant pressure and T is the temperature.

(A4) The heat flux per area is given by $\boldsymbol{q} = -\kappa \nabla T$ (Fourier's law), where κ denotes the thermal conductivity.

(A5) The mass density ϱ and the specific heat capacity c_P do not change over time.

(A6) The thermal conductivity κ is a constant in space.

These assumptions lead to the classical heat conduction equation given by

$$\varrho c_P \frac{\partial T}{\partial t} = \kappa \Delta T, \text{ or} \tag{1}$$

$$\frac{\partial T}{\partial t} = \alpha \Delta T \text{ in } \Omega, \tag{2}$$

with the thermal diffusivity $\alpha = \kappa / c_P / \varrho$. The following table assigns the physical units to the quantities just introduced.

Symbol	**SI Unit**	**Description**	**Typical Value**
c_p	$\frac{\mathrm{J}}{\mathrm{kg}\cdot\mathrm{K}}$	Heat capacity at constant pressure	4181
κ	$\frac{\mathrm{W}}{\mathrm{m}\cdot\mathrm{K}}$	Heat conductivity	0.6
α	$\frac{\mathrm{m}^{d-1}}{\mathrm{s}}$	Thermal diffusivity	1.438 e-7

Table 2: Physical quantities.

General Assumption: From now on we assume that Ω is a 2D quadratic domain $\Omega = (0,1) \times (0,1)$. Then the classical heat equation can be written as

$$\frac{\partial T}{\partial t} = \alpha \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} \right) + f = \alpha \Delta T + f. \tag{3}$$

1.2 Initial and Boundary Conditions

For a complete well-posed the heat conduction problem (3) has to be completed with initial and boundary values. Robin boundary conditions cover Dirichlet and Neumann boundary conditions, i.e.

$$aT + b \frac{\partial T}{\partial \nu} = c, \text{ or } \gamma (T - T_b) + \frac{\partial T}{\partial \nu} = 0, \text{ on } \partial\Omega \times (0, T)$$

for solutions T of $\partial T / \partial t = \alpha \Delta T + f$ and $aT_b = c, \ \gamma = a/b$.

Remark 1.1. From now on we *always assume* $\gamma > 0$. That is important to prove some of the essential results in the following section.

A typical initial condition which will be used here is $T(x, y, 0) = T_0(x, y)$.

2 Eigenfunctions and Green's Function

The idea of Green's function is to provide an exact solution which fulfills (3) subject to initial and boundary condition. For simple domains Ω an analytical expression can be calculated which can be used to calculate values of T at every spatial point $(x, y) \in \Omega$ to any time $t > 0$. For difficult domains it is not possible to derive a useful analytical expression for the Green's function that allows for value evaluations of T. Nevertheless Green's function allow to get insights and estimates for a solution. Since the solution of a well-posed problem is unique, insights, e.g. a-priori estimates and regularity results from Green's function are valid for the solution of the problem no matter in which way this solution has been calculated.

2.1 The homogeneous Heat Conduction Problem

To construct a Green's function first a particular solution of the homogenous problem, i.e.

$$\begin{aligned} \frac{\partial w}{\partial t} &= \alpha\Delta w, \text{ in } \Omega, && (4)\\ w(x,y,0) &= w_0(x,y), && (5)\\ \gamma w + \frac{\partial w}{\partial \nu} &= 0 \text{ on } \partial\Omega. && (6) \end{aligned}$$

has to be constructed. For this purpose we closely follow [10, p. 50 f].

First we apply a product approach $w(x, y, t) = \Psi(x, y) \cdot \Theta(t)$ for w. Then (4) turns into $\Psi(x, y) \cdot \Theta'(t) = \alpha\Delta\Psi(x, y) \cdot \Theta(t)$ which can be rewritten as

$$\frac{\Theta'(t)}{\alpha\Theta(t)} = \frac{\Delta\Psi(x,y)}{\Psi(x,y)} = -\Lambda. \tag{7}$$

Equation (7) can be fulfilled, only if every quotient therein is constant and hence Λ is a constant. This leads to two eigenvalue problems:

$$\Theta'(t) = -\alpha\Lambda\Theta(t), \text{ and} \tag{8}$$

$$\Delta\Psi(x,y) = -\Lambda\Psi(x,y). \tag{9}$$

We skip the case $\Lambda < 0$ and classical solution theory of linear ordinary differential equations leads to the solution $\Theta(t) = A_0\text{exp}(-\alpha\Lambda(t - t_0))$ of (8). With $t_0 = 0$ this is $\Theta(t) = A_0\text{exp}(-\alpha\Lambda t)$ for $\Lambda > 0$. For $\Lambda = 0$ the equation (8) is solved by a constant.

We are now going to solve (9). Again we adopt a product approach $\Psi(x, y) = X(x) \cdot Y(y)$ which is reasonable for a cartesian-like domain $\Omega = (0, 1) \times (0, 1)$. Then (9) turns into $X''(x)Y(y) + X(x)Y''(y) = -\Lambda X(x)Y(y)$, or

$$\frac{X''(x)}{X(x)} + \frac{Y''(y)}{Y(y)} = -\Lambda. \tag{10}$$

Every summand in (10) must be a constant, thus

$$\frac{X''(x)}{X(x)} = -\mu^2, \text{ and } \frac{Y''(y)}{Y(y)} = -\lambda^2, \text{ and } \mu^2 + \lambda^2 = \Lambda, \tag{11}$$

with constants μ and λ. The equations (11) describe two Sturm-Liouville eigenvalue problems, cf. [16, p. 285], precisely we have

$$X''(x) + \mu^2 X(x) = 0 \text{ and } \gamma X(0) - X'(0) = 0 \text{ and } \gamma X(1) + X'(1) = 0, \tag{12}$$

for $X = X(x)$, and

$$Y''(y) + \lambda^2 Y(y) = 0 \text{ and } \gamma Y(0) - Y'(0) = 0 \text{ and } \gamma Y(1) + Y'(1) = 0, \tag{13}$$

for $Y = Y(y)$. We now a state a first result, that is

Lemma 2.1 (Eigenvalues of the Sturm-Liouville problem)**.** *For the Sturm-Liouville problems (12) and (13) there exist infinitely many eigenvalues μ^2 and λ^2. For the sequences of eigenvalues $\{\mu_n^2\}_{n\geq 0}$ of (12) and $\{\lambda_n^2\}_{n\geq 0}$ of (13) we have*

$$0 \leq \mu_0^2 < \dots < \mu_m^2 < \mu_{m+1}^2 < \dots \text{ and } 0 \leq \lambda_0^2 < \dots < \lambda_n^2 < \lambda_{n+1}^2 < \dots, \tag{14}$$

for all $m, n \in \mathbb{N}$. Furthermore, the sequences have the (improper) limits

$$\lim_{n\to\infty} \mu_n^2 = +\infty \text{ and } \lim_{n\to\infty} \lambda_n^2 = +\infty. \tag{15}$$

For the proof we use techniques of the spectral theory of linear operators. For brevity we cannot go in detail here. Essentials on spectral theory can be found in every good textbook on functional analysis, for example in [10, pp 130 f], or in [7, pp 465–505, pp 542–571].

Definition 2.1 (Spectrum and resolvent set)**.** For a linear operator $A \in \mathscr{L}(H)$ on a Hilbert space H we define the *resolvent set* $\varrho(A)$ as the subset of the complex plane $\mathbb{C}$ containing all $z \in \mathbb{C}$ for which $A - zI$ has a continuous inverse $(A - zI)^{-1} \in \mathscr{L}(H)$. The complement set $\sigma(A) = \mathbb{C} \setminus \varrho(A)$ is called the *spectrum* of A. The operator $R(z, A) = (A - zI)^{-1}$ is called *resolvent*.

Proof. We define a linear operator $A : H^2(\Omega) \to L^2(\Omega)$ via the sesquilinear functional $\ell : H^2(\Omega) \times H^2(\Omega) \to \mathbb{C}$ given by the expression

$$\begin{aligned} \ell(w, \xi) &= (Aw, \xi)_{L^2(\Omega)} & (16)\\ &= (\nabla w, \nabla \xi)_{L^2(\Omega,\mathbb{R}^d)} + \gamma\, (w, \xi)_{L^2(\partial\Omega,\mathbb{R})} & (17)\\ &= \int_\Omega \nabla w \cdot \nabla \xi \mathrm{d}\mu + \gamma \int_{\partial\Omega} w\xi \mathrm{d}S. & (18)\end{aligned}$$

Note that we could have chosen $H^1(\Omega)$ instead of $H^2(\Omega)$. By definition, the operator A is self-adjoint and for all $z \in \mathbb{C}$. The operator $A - zI$ maps $H^2(\Omega)$ into $L^2(\Omega)$ as A and is self-adjoint too. Then, its resolvent $R(z, A) = (A - zI)^{-1}$ for $z \in \varrho(A)$ is a linear mapping from $L^2(\Omega)$ to $H^2(\Omega)$. According to standard theory of Sobolev spaces the embedding $j : H^2(\Omega) \to L^2(\Omega)$ is a compact operator, hence $j \circ R(z, A) : L^2(\Omega) \to L^2(\Omega)$ is compact, too. Redefining $R(z, A) = j \circ R(z, A)$ the resolvent identity $R(z, A) - R(z_0, A) = (z_0 - z)R(z, A)R(z_0, A)$ assures compact $R(z, A)$ for all $z \in \varrho(A)$. Next we apply [10, Theorem 15.12, p 137] that allows to write

$$R(z, A)\psi_n = \zeta_n \psi_n \text{ and } \psi_n = \zeta_n (A - zI)\psi_n \text{ for all } z \in \varrho(A)$$

with a sequence $\{\zeta_n\}_{n\in\mathbb{N}}$ of complex numbers of $\sigma(R(z, A))$ tending to zero as n tends to infinity and eigenfunctions ψ_n attached to ζ_n. Vice versa, $z_n = 1/\zeta_n$ define the eigenvalues of $(A - zI)$, i.e.

$$(A - zI)\psi_n = z_n \psi_n \text{ and } A\psi_n = \eta_n \psi_n,$$

where $\eta_n = z + z_n$. Since ζ_n has zero limit η_n tends to infinity as $n \to \infty$. Note that $\eta_n \in \sigma(A)$ and since A is self-adjoint we have $\eta_n \in \sigma(A) \subset \mathbb{R}$. Inserting $w = \psi_n$ and $\xi = \psi_n$ into (16) we have

$$\int_\Omega |\nabla \psi_n|^2 \, \mathrm{d}\mu + \gamma \int_{\partial\Omega} |\psi_n|^2 \, \mathrm{d}S = \eta_n \int_\Omega |\psi_n|^2 \, \mathrm{d}\mu \geq 0,$$

that is $\eta_n \geq 0$ and $\eta_n = \mu_n^2$ is justified for $\mu_n \in \mathbb{R}$. The proof of (14) and (15) is done. $\square$

Remark 2.1. Later we will see that all eigenvalues are non-zero and positive in case of proper Robin boundary conditions.

Remark 2.2. In [16, p. 286] there is an alternative proof of Lemma 2.1 using just elementary theory of ordinary differential equations and avoiding deep spectral theory of function spaces.

The solution of (11) is given by

$$X(x) = \begin{cases} A_1 x + A_2, & \text{if } \mu = 0, \\ B_1 \sin(\mu x) + B_2 \cos(\mu x), & \text{otherwise.} \end{cases} \tag{19}$$

We are now going to match the boundary conditions (6) for the solution X. For $w(x, y, t) = X(x) \cdot Y(y) \cdot \Theta(t)$ (6) reduces to

$$\gamma X(x) + \frac{\partial X(x)}{\partial \nu} = 0 \text{ on } \{0, 1\} \times (0, 1).$$

The normal derivative $\partial X/\partial\nu$ on $\{0, 1\} \times (0, 1)$ is given by

$$\frac{\partial X(x)}{\partial \nu} = \begin{cases} -X'(0) & \text{on } \{0\} \times (0, 1), \\ X'(1) & \text{on } \{1\} \times (0, 1), \end{cases}$$

and the boundary condition as

$$0 = \gamma X(x) + \frac{\partial X(x)}{\partial \nu} = \begin{cases} \gamma B_2 - \mu B_1 & \text{for } \mu \neq 0 \text{ on } \{0\} \times (0, 1), \\ (\gamma B_1 - \mu B_2)\sin(\mu) + \dots & \\ \dots + (\gamma B_2 + \mu B_1)\cos(\mu) & \text{for } \mu \neq 0 \text{ on } \{1\} \times (0, 1), \\ \gamma A_2 - A_1 & \text{for } \mu = 0 \text{ on } \{0\} \times (0, 1), \\ \gamma(A_1 + A_2) + A_1 & \text{for } \mu = 0 \text{ on } \{1\} \times (0, 1). \end{cases} \tag{20}$$

For $\mu \neq 0$ the first two cases of (20) can be combined to give

$$F(\mu) = B_1 \cdot \Big((\gamma^2 - \mu^2) \sin(\mu) + 2\mu\gamma \cos(\mu) \Big) = 0. \tag{21}$$

The term of the left hand side of (21) is a function F of μ. Then the zeros of F are solutions of (21). The following chart shows this function F and its first zeros for $\gamma = 1/2$.

Function $F(\mu) = (\gamma^2 - \mu^2)\sin(\mu) + 2\mu\gamma\cos(\mu)$ and its Zeros

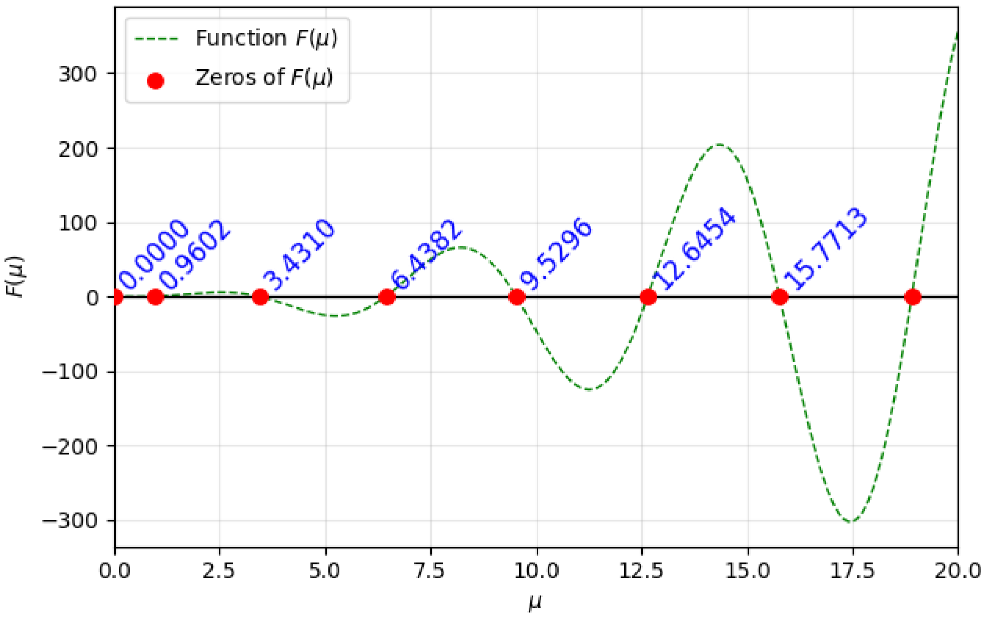


Figure 1: Function $F(\mu)$ and its zeros.

As one notices, $\mu = 0$ is a zero of $F(\mu)$. Thus there is the question, if $\mu = 0$ could be admissible for the approach $X(x) = B_1 \sin(\mu x) + B_2 \cos(\mu x)$. In this case X reduces to $X(x) = B_2 \cos(\mu x) = B_2$ and this inserted into (6) is $\gamma B_2 + 0 = 0$ which is $B_2 = 0$ and $X(x) = 0$ whenever $\gamma \neq 0$. A similar consideration leads to the conclusion that the approach $X(x) = A_1 x + A_2$ leads to $A_1 = A_2 = 0$ and again to $X(x) = 0$. But $X(x) = 0$ cannot serve as an eigenelement. Thus, we have proved the following theorem:

Theorem 2.1 (Eigenvalues are non-zero for $\gamma \neq 0$)**.** *In case of proper Robin boundary conditions, i.e. $\gamma \neq 0$ the value $\mu = 0$ is not an element of the spectrum, i.e. it is* not *an eigenvalue.*

For $\gamma = 0$ which indicates homogeneous Neumann boundary conditions (21) reduces to $\mu^2 B_1 \sin(\mu) = 0$. Since μ was assumed to be non-zero, only $B_1 = 0$ is possible and $X(x) = B_2 \cos(\mu x)$. Then, from the second case of (20) follows $\mu \in \{n \cdot \pi \text{ for all } n \in \mathbb{N} \cup \{0\}\}$.

An observation of the first eigenvalues for the case $\gamma \neq 0$ may lead to the assumption that each of the

	Neumann BC	**Robin BC**
γ	0	non-zero
μ	$n\pi, n \in \mathbb{N} \cup \{0\}$	$\{\mu : (\gamma^2 - \mu^2)\sin(\mu) + 2\mu\gamma\cos(\mu) = 0\} \setminus \{0\}$
1st eigenvalues μ^2	0, π^2, $4\pi^2$	0.921963, 11.771859, 41.450383

Table 3: Comparison of the different boundary conditions.

eigenvalues $\{6.4382,\ 9.5264,\ 12.6454,\ 15.7713, \dots\}$ have a distance of approximately π to its nearest neighbor. The following result provides a lower bound for the eigenvalues.

Lemma 2.2 (Lower estimate for the eigenvalues)**.** *In case of $\gamma \neq 0$ the eigenvalues μ_n with $\mu_n > \gamma$ fulfill $\mu_n \geq n \cdot \pi$, if $\mu_0 = 0.9602$.*

Proof. Consider the equation

$$\left(\gamma^2 - \mu^2\right) \cdot \tan(\mu) + 2\mu\gamma = 0, \text{ or } \tan(\mu) = -\frac{2\mu\gamma}{(\gamma^2 - \mu^2)}.$$

The solutions are given by

$$\mu_n = n \cdot \pi + \arctan\left(\frac{2\mu_n\gamma}{(\gamma^2 - \mu_n^2)}\right).$$

Since $\gamma/\mu_n < 1$ we can use the addition theorem for the arc tangent, cf. [3, p. 237] to get

$$\mu_n = n \cdot \pi + 2\arctan\left(\frac{\gamma}{\mu_n}\right) \geq n \cdot \pi,$$

which proves the assertion. □

In case of $\gamma \neq 0$ the eigenfunctions are given by

$$\phi_n(x) = C_n \cdot \left(\sin(\mu_n x) + \frac{\mu_n}{\gamma}\cos(\mu_n x)\right) \text{ where } n > 0, \tag{22}$$

and C_n so that $\|\phi_n\|_{L^2} = 1$. In detail, C_n fulfills

$$C_n = \left(\frac{1}{2} - \frac{\sin(2\mu_n)}{4\mu_n} + \frac{1}{2\gamma}(1 - \cos(2\mu_n)) + \frac{\mu_n^2}{\gamma^2}\left(\frac{1}{2} + \frac{\sin(2\mu_n)}{4\mu_n}\right)\right)^{-\frac{1}{2}}. \tag{23}$$

In case of $\gamma = 0$ the eigenfunctions are given by

$$\phi_0(x) = 1 \text{ and } \phi_n(x) = D_n \cdot \cos(\mu_n x) \text{ where } n > 0, \tag{24}$$

and $D_n = \sqrt{2/\mu_n}$.

Since $-\Delta : H_1 \to H_2$ is a self-adjoint linear operator between two appropriate Hilbert spaces H_1 and H_2 the systems (22) and (24) form an orthonormal basis of H_1. A possible and safe choice for the Hilbert spaces is $H_1 = H^{2,2}(0,1)$ and $H_2 = L^2(0,1)$. As x varies in $(0,1)$ so y does and the consideration done for x can analogously applied to the y variable. So for $\gamma \neq 0$ we have

$$\phi_n(y) = C_n \cdot \left(\sin(\mu_n y) + \frac{\mu_n}{\gamma}\cos(\mu_n y)\right) \text{ where } n > 0, \tag{25}$$

with the same numbers μ_n and C_n.

Theorem 2.2 (Uniform boundedness of the eigenfunctions ϕ_n)**.** *For the eigenfunctions ϕ_n there is*

$$\sup_{x\in(0,1)}|\phi_n(x)| \leq \sqrt{1.1 + 0.3\gamma^2},$$

if the spectrum does not contain zero.

Proof. For an arbitrarily fixed n define $z = 2\mu_n$, and $\beta = 2\gamma$, just for simplicity. Now, let us define the following Laurent polynomial $p(z) = C_{-1}\frac{1}{z} + p_0(z)$, where $p_0(z) = C_0 + C_1 z + C_2 z^2$, and

$$\begin{aligned} C_{-1} &= -\sin(z), \\ C_0 &= 1 + \frac{2}{\beta}(1 - \cos(z)), \\ C_1 &= \frac{\sin(z)}{\beta^2}, \\ C_2 &= \frac{1}{\beta^2}. \end{aligned}$$

The polynomial p_0 can be estimated as follows

$$\begin{aligned} C_0 &\geq 1, \\ C_1 z + C_2 z^2 &\geq \frac{z}{\beta^2}(z-1), \end{aligned}$$

that is $p_0(z) \geq 1 + \beta^{-2}z(z-1)$. It remains to estimate $C_{-1}z^{-1}$. In case of homogeneous Neumann boundary conditions the spectrum of the Laplace operator contains the eigenvalue $\mu = z = 0$ and we cannot estimate anything. In contrast in case of Robin boundary conditions, its spectrum does not contain zero. So, we may assume $z > 2\mu_1 = 2 \cdot 0.9602... = 1.9204$ which is the double of the first positive zero of F depicted in Figure 1. With $\sin(2\mu_1)/(2\mu_1) \approx 0.489225$ we get $C_{-1}/z \geq -0.489225$. Collecting all together we finally get $p(z) \geq 1 + \beta^{-2}z(z-1) - 0.489225 \geq 1/2 + \beta^{-2}z(z-1)$. Note that $2C_n^2 = 1/p_0(2\mu_n)$ and thus $C_n \leq 1/2 \cdot (1/2 + \beta^{-2}(2\mu_n)((2\mu_n) - 1))^{-1/2}$. Then we have for the n-th eigenfunction the following estimate

$$\begin{aligned}
(\phi_n(x))^2 &= 2C_n^2 \cdot \left(\sin(\mu_n x) + \frac{\mu_n}{\gamma}\cos(\mu_n x)\right)^2 \\
&\leq \frac{\left(\sin(\mu_n x) + \frac{\mu_n}{\gamma}\cos(\mu_n x)\right)^2}{\frac{1}{2} + (2\gamma^{-2})\mu_n(\mu_n - 1)} \\
&\leq \frac{\left(1 + \frac{\mu_n}{\gamma}\right)^2}{(2\gamma)^{-2}\mu_n(\mu_n - 1)} \leq 2\frac{1 + \left(\frac{\mu_n}{\gamma}\right)^2}{(2\gamma)^{-2}\mu_n(\mu_n - 1)} \\
&\leq \gamma^2\frac{1 + \left(\frac{\mu_n}{\gamma}\right)^2}{2\mu_n(\mu_n - 1)} = \frac{\gamma^2 + \mu_n^2}{2\mu_n(\mu_n - 1)} \\
&= \frac{(\mu_n^2 - \mu_n) + (\mu_n + \gamma^2)}{2\mu_n(\mu_n - 1)} \\
&= \frac{1}{2} + \frac{1}{2\mu_n - 1} + \frac{\gamma^2}{2\mu_n(\mu_n - 1)} \\
&\leq \frac{1}{2} + \frac{1}{2 \cdot 0.9204} + \frac{\gamma^2}{2 \cdot 1.9204 \cdot 0.9204} \\
&\leq 0.5 + 0.543... + \frac{\gamma^2}{3.5417...} \leq 1.1 + 0.3\gamma^2 \\
&\leq C,
\end{aligned}$$

that is $|\phi_n(x)| \leq \sqrt{1.1 + 0.3\gamma^2}$, uniformly for all $n \in \mathbb{N}$. □

Defining $\phi_{m,n}(x,y) = \phi_m(x) \cdot \phi_n(y)$ we have an orthonormal basis of $H^{2,2}(\Omega) = H^{2,2}(0,1) \times H^{2,2}(0,1)$. Then, every function $f \in H^{2,2}(\Omega)$ can be written as $f = \sum_{m,n \in I \times I} (\phi_{m,n}, f)_{L^2}\phi_{m,n}$, where the index set $I = \mathbb{N}$ in case of $\gamma \neq 0$, and $I = \mathbb{N} \cup \{0\}$ in case of $\gamma = 0$.

Theorem 2.3 (Uniform boundedness of the eigenfunctions $\phi_{m,n}$)**.** *For the eigenfunctions $\phi_{m,n} : \Omega = (0,1)^2 \to \mathbb{R}$ there is*

$$\sup_{(x,y)\in\Omega}|\phi_{m,n}(x,y)| \leq 1.1 + 0.3\gamma^2,$$

if Robin boundary conditions are imposed on the boundary.

Proof. This is a direct consequence of the Theorem 2.2. □

We are now define Green's function as

$$\mathscr{G}(x,y,x',y',\tau) = H(\tau) \sum_{m,n \in I \times I} \phi_{m,n}(x,y)\phi_{m,n}(x',y') \cdot \exp(-\alpha(\mu_m^2 + \lambda_n^2)\tau), \tag{26}$$

where H is the Heaviside function which is one for $\tau > 0$ and zero otherwise. With the help of this function a solution w of (4, 5, 6) can be constructed as

$$w(x,y,t) = \int_0^1 \int_0^1 \mathscr{G}(x,y,x',y',t) w_0(x',y') \mathrm{d}x' \mathrm{d}y'. \tag{27}$$

Since $\Delta\mathscr{G} = \delta$ in the sense of distributions adding a term $\int_0^t \int_0^1 \int_0^1 \mathscr{G}(x,y,x',y',t-t') f(x',y') \mathrm{d}x' \mathrm{d}y' \mathrm{d}t'$ to (2.1) allows (4) to be extended by the function f. A solution w of this inhomogenized problem is then given by

$$w(x,y,t) = \int_0^1\int_0^1 \mathscr{G}(x,y,x',y',t)w_0(x',y')\mathrm{d}x'\mathrm{d}y' \tag{28}$$
$$+ \int_0^t\int_0^1\int_0^1 \mathscr{G}(x,y,x',y',t-t')f(x',y')\mathrm{d}x'\mathrm{d}y'\mathrm{d}t'. \tag{29}$$

Theorem 2.4 (Long-time stability of solutions w)**.** *The solution $w(x,y,t)$ of the homogeneous problem is uniformly bounded for all $(x,y,t)\in(0,1)^2\times(0,\infty)$, if Robin boundary conditions are imposed on the boundary.*

Proof. We consider the integral

$$I_N(x,y,t) = \int_0^t\int_0^1\int_0^1 \sum_{m,n=1}^{N} \phi_{m,n}(x,y)\phi_{m,n}(x',y')\cdot f(x',y')a_{m,n}(\tau)\mathrm{d}x'\mathrm{d}y'\mathrm{d}\tau,$$

with $a_{m,n}(\tau)=H(\tau)\exp(-\alpha(\mu_m^2+\lambda_n^2)\tau)$ as amplitude function. Since $\sum_{m,n=1}^N\phi_{m,n}(x,y)\phi_{m,n}(x',y')\cdot f(x',y')$ does not depend on τ the integration may be decoupled. We then have

$$\int_0^t a_{m,n}(\tau)\mathrm{d}\tau = \frac{1-\mathrm{e}^{-\alpha(\mu_m^2+\lambda_n^2)t}}{\alpha(\mu_m^2+\lambda_n^2)} \le \frac{1}{\alpha(\mu_m^2+\lambda_n^2)} \le \frac{1}{2\alpha\mu_1^2}.$$

Furthermore, using Theorem 2.3 we get

$$\begin{aligned}|I_N(x,y,t)| &= \left|\sum_{m,n=1}^{N}\frac{1-\mathrm{e}^{-\alpha(\mu_m^2+\lambda_n^2)t}}{\alpha(\mu_m^2+\lambda_n^2)}\int_0^1\int_0^1\phi_{m,n}(x,y)\phi_{m,n}(x',y')\cdot f(x',y')\mathrm{d}x'\mathrm{d}y'\right|\\ &\le \left(1.1+0.3\gamma^2\right)\left|\sum_{m,n=1}^{N}\frac{1-\mathrm{e}^{-\alpha(\mu_m^2+\lambda_n^2)t}}{\alpha(\mu_m^2+\lambda_n^2)}\int_0^1\int_0^1\phi_{m,n}(x',y')\cdot f(x',y')\mathrm{d}x'\mathrm{d}y'\right|.\end{aligned}$$

Now, defining $c_{m,n}=\int_0^1\int_0^1\phi_{m,n}(x',y')\cdot f(x',y')\mathrm{d}x'\mathrm{d}y'$ note that $\{c_{m,n}\}_{m,n\in\mathbb{N}}$ is a sequence in the sequence space ℓ^2, that means that $\{|c_{m,n}|\}_{m,n\in\mathbb{N}}$ is bounded by a constant $d_1\in\mathbb{R}$. Without loss of generality, see Lemma 2.1 we may assume that $\mu_m^2>\gamma$ and $\lambda_n^2>\gamma$. We then have

$$\begin{aligned}\sum_{m,n=1}^{N}\frac{|c_{m,n}|}{\alpha(\mu_m^2+\lambda_n^2)} &\le d_1\sum_{m,n=1}^{N}\frac{1}{\alpha(\mu_m^2+\lambda_n^2)}\\ &\le d_1\sum_{m,n=1}^{N}\frac{1}{\alpha(m^2+n^2)},\end{aligned}$$

the last inequality as a consequence of Lemma 2.2. Furthermore, the right hand side of the last inequality is the partial sum of an absolutely convergent series, i.e. $\{1/(m^2+n^2)\} \in \ell^1$ with limit d_2, and

$$
\begin{aligned}
& \left| \sum_{m,n=1}^{N} \frac{1}{\alpha(\mu_m^2+\lambda_n^2)} \int_0^1 \int_0^1 \phi_{m,n}(x',y') \cdot f(x',y') \mathrm{d}x' \mathrm{d}y' \right| \\
\leq & \sum_{m,n=1}^{N} \frac{1}{\alpha(\mu_m^2+\lambda_n^2)} \left| \int_0^1 \int_0^1 \phi_{m,n}(x',y') \cdot f(x',y') \mathrm{d}x' \mathrm{d}y' \right| \\
\leq & \sum_{m,n=1}^{N} \frac{1}{\alpha(m^2+n^2)} \left| \int_0^1 \int_0^1 \phi_{m,n}(x',y') \cdot f(x',y') \mathrm{d}x' \mathrm{d}y' \right| \\
\leq & \ d_1 \cdot d_2 \in \mathbb{R}.
\end{aligned}
$$

Since $1 - \mathrm{e}^{-\alpha(\mu_m^2+\lambda_n^2)t} \leq 1$ we finally have

$$
|I_N(x,y,t)| \leq \left(1.1 + 0.3\gamma^2\right) d_1 \cdot d_2,
$$

uniformly for all N and all $(x,y) \in (0,1)^2$. That means that the solution $w(x,y,t)$ of the homogeneous problem is uniformly bounded for all $(x,y,t) \in (0,1)^2 \times (0,\infty)$, the statement of the theorem. □

Remark 2.3. If the homogeneous heat conduction problem is supplemented with homogeneous Neumann boundary conditions the assertion of Theorem 2.4 is not true: It can be shown that the solutions w will tend to infinity as the time t goes to infinity.

2.2 The inhomogeneous Heat Conduction Problem

The full inhomogenous heat conduction problem is given by

$$
\begin{aligned}
\frac{\partial u}{\partial t} &= \alpha \Delta u + f, \text{ in } \Omega, && (30) \\
u(x,y,0) &= u_0(x,y), && (31) \\
\gamma(u - u_{\mathrm{amb}}) + \frac{\partial u}{\partial \nu} &= 0 \text{ on } \partial\Omega. && (32)
\end{aligned}
$$

We assume u_{amb} is a constant. With $w = u - u_{\mathrm{amb}}$ and $w_0 = u_0 - u_{\mathrm{amb}}$ a solution of the problem $\partial w/\partial t = \alpha \Delta w + f$ and the conditions (5, 6) can be constructed via (28, 29). Vice versa, a solution of the inhomogeneous problem (30, 31, 32) can be defined via $u = w + u_{\mathrm{amb}}$ where w is constructed by (28, 29) again.

We finish this section with two important result, one for the case of no-flux boundary conditions, the other for the case of Robin boundary conditions:

Theorem 2.5 (Long-term behavior of the solution of (3) in case of no-flux boundary conditions)**.** *With no-flux boundary conditions, i.e. zero normal derivatives of u at the boundary of Ω and time-constant bulk heating by internal heat sources f solutions u of the heat equation (3) will attain inifinite large temperatures when the time t tends to zero.*

Proof. We consider the time-space integrated version of (3), i.e.

$$\int_0^t \int_\Omega \frac{\partial u}{\partial t} \mathrm{d}\mu \mathrm{d}t = \alpha \int_0^t \int_\Omega \Delta u \mathrm{d}\mu \mathrm{d}t + \int_0^t \int_\Omega f \mathrm{d}\mu \mathrm{d}t.$$

Modifying the left hand side we have

$$\begin{aligned} \int_0^t \int_\Omega \frac{\partial u}{\partial t} \mathrm{d}\mu \mathrm{d}t &= \int_\Omega \int_0^t \frac{\partial u}{\partial t} \mathrm{d}t \mathrm{d}\mu \\ &= \int_\Omega u(t) \mathrm{d}\mu - \int_\Omega u(0) \mathrm{d}\mu. \end{aligned}$$

Since f does not depend upon time t we obtain

$$\begin{aligned} \int_\Omega u(t) \mathrm{d}\mu &= \int_\Omega u(0) \mathrm{d}\mu + \alpha \int_0^t \int_\Omega \Delta u \mathrm{d}\mu \mathrm{d}t + \int_0^t \int_\Omega f \mathrm{d}\mu \mathrm{d}t \\ &= \int_\Omega u(0) \mathrm{d}\mu + \alpha \int_0^t \int_{\partial\Omega} \frac{\partial u}{\partial \nu_{\partial\Omega}} \mathrm{d}S \mathrm{d}t + t \int_\Omega f \mathrm{d}\mu, \end{aligned}$$

where we have applied the Gauss-Green theorem, see [14, p. 288] and [4, p.312], on $\int_0^t \int_\Omega \Delta u \mathrm{d}\mu \mathrm{d}t$. Now, using the no-flux boundary condition we have

$$\begin{aligned} \sup_{(x,y)\in\Omega} |u(x,y)| &\geq \frac{1}{\mu(\Omega)} \int_\Omega u(t) \mathrm{d}\mu \\ &\geq \frac{\alpha}{\mu(\Omega)} \int_0^t \int_{\partial\Omega} \frac{\partial u}{\partial \nu_{\partial\Omega}} \mathrm{d}S \mathrm{d}t + \frac{t}{\mu(\Omega)} \int_\Omega f \mathrm{d}\mu \\ &\geq \frac{\alpha}{\mu(\Omega)} \int_0^t \int_{\partial\Omega} 0 \ \mathrm{d}S \mathrm{d}t + \mathrm{const} \cdot t \\ &\geq \mathrm{const} \cdot t, \end{aligned}$$

that is $\|u\|_{L^\infty(\Omega)} = \sup_{(x,y)\in\Omega} |u(x,y,t)|$ tends to infinity as $t \to \infty$. □

The following important variant of Theorem 2.4 is straightforward:

Theorem 2.6 (Long-time stability of solutions u of (30, 31, 32))**.** *The solution $u(x,y,t)$ of the inhomogeneous problem (30, 31, 32) is uniformly bounded for all $(x,y,t) \in (0,1)^2 \times (0,\infty)$.*

3 Finite Difference Method

The ideas of finite difference method is to approximate spatial derivatives by numerical difference quotients. Evaluating a difference quotients at a certain point (x,y) requires its neighboring points $(x-\Delta x, y)$, $(x+\Delta x, y)$, $(x, y-\Delta y)$ and $(x, y+\Delta y)$. For this purpose a set of grid points $\{(x_i, y_j)\|(i,j) \in I\}$ in Ω. Since $\Omega = (0,1)\times(0,1)$ the grid points $\{(x_i, y_j)$ are aligned with the coordinate axes. Setting $h = \Delta x = \Delta y$ we approximate the second order spatial derivatives by

$$\begin{aligned} \Delta v(x,y) &\approx \Delta_h v(x,y) \\ &= \frac{1}{h^2} \left(v(x+h,y) - 2v(x,y) + v(x-h,y) \right) \\ &+ \frac{1}{h^2} \left(v(x,y+h) - 2v(x,y) + v(x,y-h) \right), \end{aligned}$$

and the first order time derivative by a forward difference

$$\frac{\mathrm{d}w(t))}{\mathrm{d}t} \approx \frac{w(t+\tau)}{\tau}.$$

We now use u as an alias for T, i.e. $u = T$. With a definite (x, y) grid and an initial state $T_0(x, y)$ at time $t = 0$ an approximation for the solution u at time $t = 0 + \tau$ can be obtained via

$$u_1 = u_0 + \alpha h \Delta_h u_0,$$

where $u_0 = T_0$ evaluated at grid points. This can be successive be done:

$$u_{k+1} = F_{\text{explicit}}(u_k) = u_k + \alpha h \Delta_h u_k,$$

what defines the explicit Euler method. In contrast the implicit method is given via

$$u_{k+1} = F_{\text{implicit}}(u_k, u_{k+1}) = u_k + \alpha h \Delta_h u_{k+1},$$

which requires to solve a system of linear equations to get u_{k+1}, but the numerical behavior of the implicit method is more robust. Combining explicit and implicit method leads to the Crank-Nicolson method as arithmetic mean of both methods

$$u_{k+1} = \tfrac{1}{2}\left(F_{\text{explicit}}(u_k) + F_{\text{implicit}}(u_k, u_{k+1})\right).$$

The stability of the explicit Euler method is limited by the CFL condition:

$$\tfrac{\tau}{h^2} \leq \tfrac{1}{4\alpha},$$

whereas for the Crank-Nicolson method

$$\tfrac{\tau}{h^2} \leq \tfrac{1}{2\alpha}$$

must be fulfilled for L^∞-stability. For more information, see [6, p 112].

4 Physics Informed Neural Networks (PINN)

Classification or *approximation* are the typical applications of neural networks. A short introduction to neural networks can be found in [15, pp. 809], [12] and [13]. Today the main applications of neural networks contain classification. On the other hand, neural networks can be used to approximate an (unknown) function. Physics informed neural networks are a special type of neural networks to solve mathematical equations of physics and engineering. Typically these equations are ordinary or partial differential equations. Whereas classical numerical solvers outperform this deep learning approach, trained PINN models can be easily reused, if, e.g. a finer spatial grid is needed or the geometry of the spatial domain varies slightly. In good cases these tasks can be done without a new training. Furthermore PINNs can be used to estimate parameters in the equation, if an approximate solution is given, e.g. by measurements. In PINNs the neural network is used as a function approximator. This approach can be found in [9, Section 3.2]. The activation functions of a neural network span a function space and a function is approximated by a linear combinations of copies of the activation functions each having different scaling and offset in its function argument. The coefficients, the scalings and offsets are simply weights of the neural network to be adjusted during training. Key ingredient of the learning procedure is the loss function which will be minimized during training. This loss function will be built up by loss terms arising from the mathematical equation to solve. That means that three loss terms will be added together to give the (total) training loss:

1. **Physics loss:** The loss arising from deviations of the approximate solution $\hat{u}$ from the exact solution u. Generally, if $F(u) = 0$ is the mathematical equation, then the physics loss is $L_{\text{Physics}}(w) = |F(w)|^2$ where w can be any admissible function. In case $w = u$ the equation $F(u) = 0$ is fulfilled and $L_{\text{Physics}}(u) = 0$ is minimal. For any approximate solution $\hat{u}$ we only have $L_{\text{Physics}}(\hat{u}) = 0$. So L_{Physics} penalizes deviations from the exact solution u.

2. **Initial loss:** Similarly, let $I(u) = 0$ denote the initial condition. Then the initial loss is defined as $L_{\text{Initial}}(w) = |I(w)|^2$.

3. **Boundary loss:** Finally, for a boundary operator $B(u) = 0$ the boundary loss is analogously defined as $L_{\text{Boundary}}(w) = |B(w)|^2$.

To set up the PINN training the approach of [1] is used and extended. The source code used therein can be found in [2]. In this thesis an inhomogeneous heat conduction equation in two spatial dimensions with Dirichlet or Neumann boundary conditions is solved using PINNs. Let us shortly summarize the key features:

1. **PINN architecture:** A feed forward neural network with five hidden layers, each of them containing 50 nodes is used. The network has input layer with three input nodes corresponding to the variables x, y, and t. The single node of the output layer is assigned to the approximate solution $\hat{u}$.

2. **PINN Activation function:** All layers except the output layer use the hyperbolic tangent as activation function.

3. **PINN Solver:** To train the PINN the Adam solver of the Pytorch package is used with default settings and a learning rate of 1e-3.

4. **Sampling:** Random samples of $N \in \{1000, 8000\}$ points, each for x, y and t, which is a Monte-Carlo sampling.

5 Case Studies

5.1 Error Measures

PDE Residual

On the spatial inner points of the numerical solution $u_h^{(n)}$ at time frame n the discrete Laplace operator Δ_h may be applied to give

$$\Delta_h u_h^{(n)}(x_i, y_j) = \frac{u_h^{(n)}(x_{i+1}, y_j) + u_h^{(n)}(x_{i-1}, y_j) + u_h^{(n)}(x_i, y_{j+1}) + u_h^{(n)}(x_i, y_{j-1}) - 4u_h^{(n)}(x_i, y_j)}{h^2},$$

for all n and all inner points (x_i, y_j). The time derivative is just

$$\frac{\partial_h u_h^{(n)}(x_i, y_j)}{\partial_h t} = \frac{u_h^{(n+1)}(x_i, y_j) - u_h^{(n-1)}(x_i, y_j)}{t_{n+1} - t_{n-1}},$$

again for all n and all inner points (x_i, y_j). A local error term now may be given by

$$E(x_i, y_j) = \left| \frac{\partial_h u_h^{(n)}(x_i, y_j)}{\partial_h t} - \alpha \Delta_h u_h^{(n)}(x_i, y_j) - f(x_i, y_j) \right|,$$

and its damped version to keep the numerics stable,

$$E_{\text{damped}}(x_i, y_j) = \left| u_h^{(n+1)}(x_i, y_j) - u_h^{(n-1)}(x_i, y_j) - (t_{n+1} - t_{n-1}) \cdot \left(\alpha \Delta_h u_h^{(n)}(x_i, y_j) + f(x_i, y_j) \right) \right|,$$

again for all n and all inner points (x_i, y_j). Possible error measures are now

1. Mean absolute error (MAE): Arithmetic mean of all local error terms $E(x_i, y_j)$.
2. Damped mean absolute error: Arithmetic mean of all local damped error terms $E_{\text{damped}}(x_i, y_j)$.
3. Mean square error (MSE): Arithmetic mean of all squared local error terms $E^2(x_i, y_j)$.
4. Root mean square error (RMSE): Square root of the MSE.

First we will use the simple MAE approach. Furthermore, the Green function as an exact solution formula allows for a closed form of the derivatives.

Boundary Residuals

1. Left boundary: $R_L(x_1, y_j) = h(u_h^{(n)}(x_1, y_j) - u_{\text{amb}}) - \frac{u_h^{(n)}(x_2, y_j) - u_h^{(n)}(x_1, y_j)}{h}$
2. Right boundary: $R_R(x_N, y_j) = h(u_h^{(n)}(x_N, y_j) - u_{\text{amb}}) + \frac{u_h^{(n)}(x_N, y_j) - u_h^{(n)}(x_{N-1}, y_j)}{h}$
3. Lower boundary (bottom): $R_B(x_i, y_1) = h(u_h^{(n)}(x_i, y_1) - u_{\text{amb}}) - \frac{u_h^{(n)}(x_i, y_2) - u_h^{(n)}(x_i, y_1)}{h}$
4. Upper boundary (top): $R_T(x_i, y_N) = h(u_h^{(n)}(x_i, y_N) - u_{\text{amb}}) + \frac{u_h^{(n)}(x_i, y_N) - u_h^{(n)}(x_i, y_{N-1})}{h}$

For each of these four boundary sections we will calculate the mean square error (MSE) to get a single error value describing the quality of the boundary match of the numerical solution on every boundary section.

5.2 Case Study One: Constant initial temperature with no-flux boundary conditions using explicit solver and PINN as solution method

This case is also discussed in [1]. With the introduces PINN setup, we use a thermal diffusivity $\alpha = 0.1$ in the heat equation. Furthermore we have a Gaussian kernel as a bulk heat source, i.e.

$$f(x, y, t) = S \cdot \exp\left(-\frac{(x^2+y^2)}{2r}\right), \tag{33}$$

with $r = 0.1$ and heat strength $S = 500.0$. As initial condition at $t = 0$ we set the temperature equal to 25 °C at every point in Ω. We assume that there is now heat flux across the boundary of Ω, i.e. $\frac{\partial u}{\partial n} = 0$ at $\partial\Omega$. For the training a Pytorch scheduler is used to adjust the learning rate adaptively. The adjustment starts after a certain number of finished epochs. This number is the step size parameter of the scheduler. We have used 4,000 and 6,000 as step size. After passing this number of epochs the learning rate decreases from 1e-3 to 5e-4 and further, each time by halving. Training with three models on the this non-homogeneous initial-boundary problem leads to the following Adjustments and results: The relative

Parameter	**Model 1**	**Model 2**	**Model 3**
Epochs	8,000	10,000	20,000
Samples	1,000	8,000	8,000
Step size	4,000	6,000	6,000
Relative error	19.03%	7.41 %	4.96 %

Table 4: Adjustment and Results.

error refers to the corresponding solution using the explicit finite-difference scheme. With more training epochs the relative error decreases.

The behaviour of the averaged temperatures shown in Figure (5) confirms the pattern stated in Theorem 2.5.

5.3 Case Study Two: Constant initial temperature with Robin boundary conditions

Some Mathematics

In case 1 the domain Ω is constantly heated over time where the heat cannot dissipate over across the boundary $\partial\Omega$ due to the no-flux boundary condition imposed on $\partial\Omega$. This leads to permanently ascending temperatures over time without reaching an equilibrium. In real life such an isolated system with permanent constant heat support is not realistic. This can be overcome by either

- using another heat source which decreases heat supply over time to zero, or
- imposing other boundary conditions which allow for heat flux over the domain boundary.

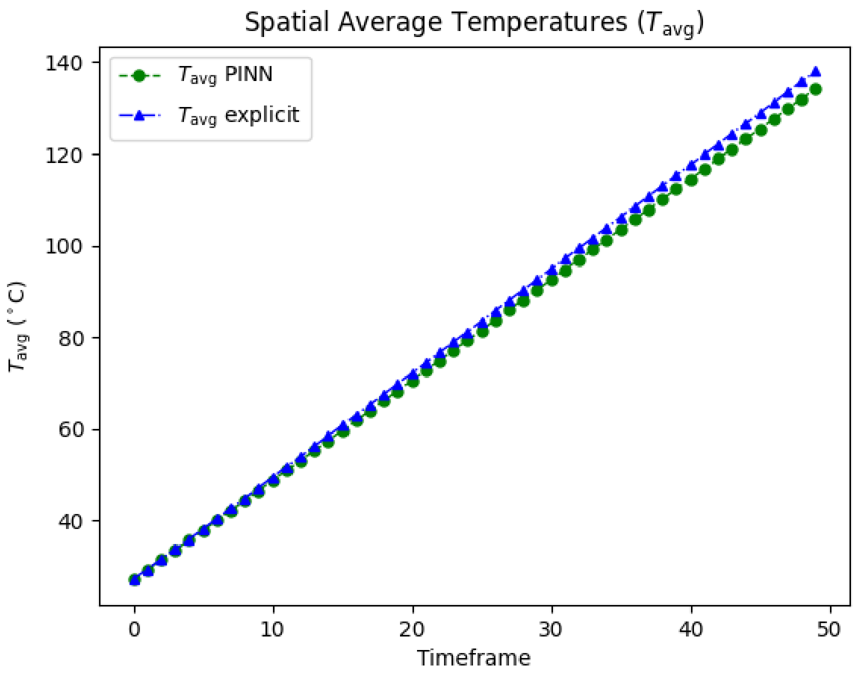
Spatial Average of the calculated Temperatures


Table 5: Temperature averages for explicit and PINN solver.

In this case the latter approach is used by imposing a Robin boundary condition, i.e.

$$a \cdot (u - u_0) + b \frac{\partial u}{\partial \nu_{\partial\Omega}} = 0, \tag{34}$$

where $\nu_{\partial\Omega}$ is the outer unit normal at the boundary $\partial\Omega$, u_0 the ambient temperature far from the domain Ω and its boundary $\partial\Omega$, i.e. far from the thermal boundary layer. The following theorem assures that the solution u of the heat equation with Robin boundary conditions will tend to a finite equlibrium state if appropriate conditions are fulfilled.

Lemma 5.1 (Positiveness)**.** *Given the Robin boundary condition*

$$hw + \frac{\partial w}{\partial \nu_{\partial\Omega}} = 0.$$

Assume Ω is the unit square, i.e. $\Omega = (0,1) \times (0,1)$, and $u(0) \geq u_0$ and $f \geq 0$ in Ω. (no heat sinks in Ω). Then $w \geq 0$ on $\partial\Omega$ holds.

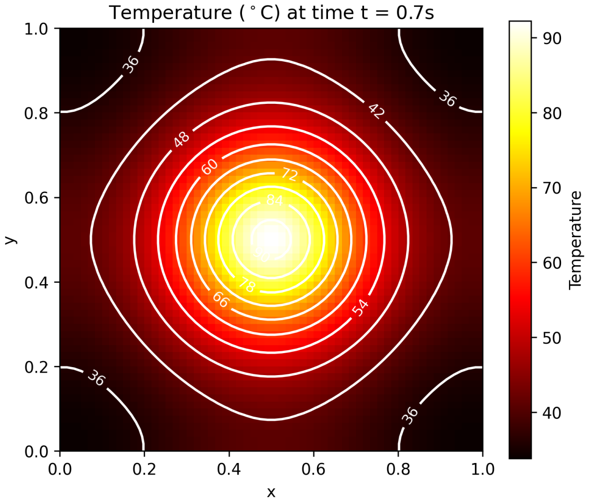
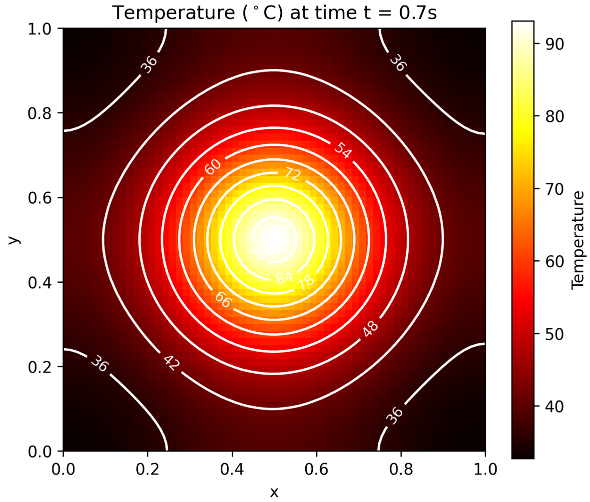
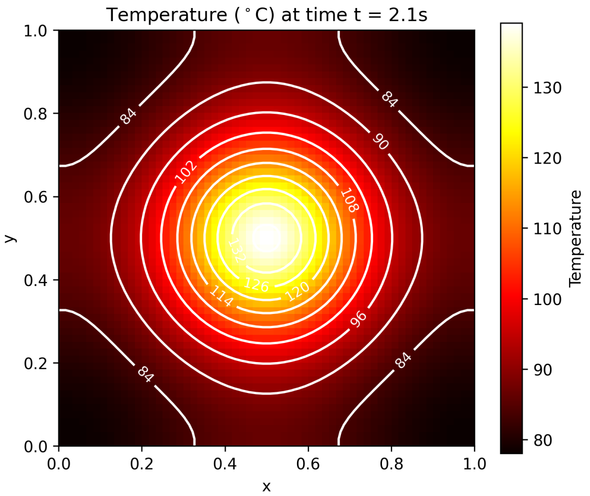
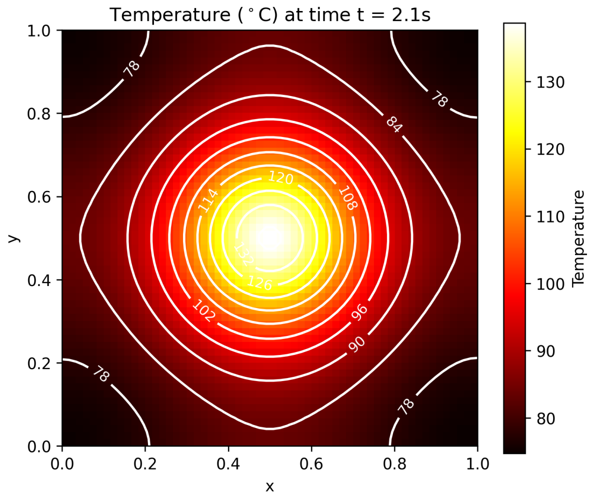
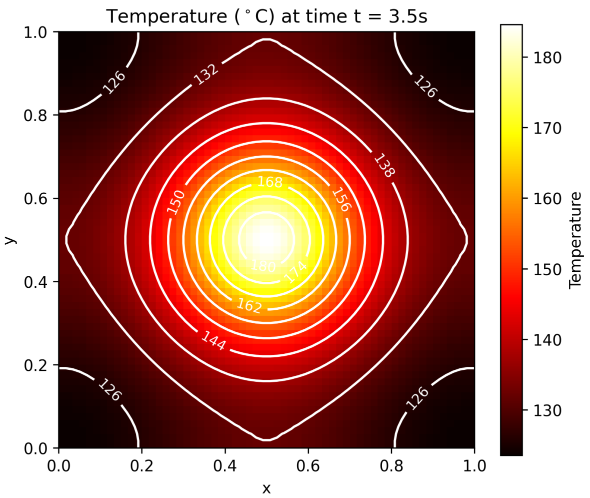
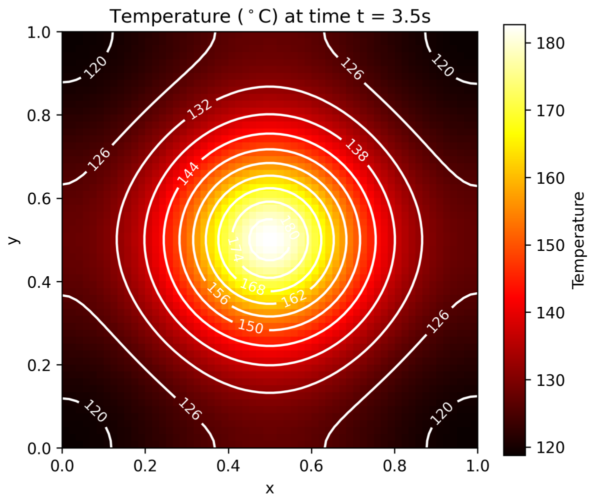
Explicit Solver	PINN
	
	
	

Table 6: Solution to different time frames.

Proof. The boundary conditions read as

$$hw + \frac{\partial w}{\partial \nu_{\partial\Omega}} = \begin{cases} hw - \frac{\partial w}{\partial x} & \text{on left boundary where } x = 0, \\ hw - \frac{\partial w}{\partial y} & \text{on lower boundary where } y = 0, \\ hw + \frac{\partial w}{\partial x} & \text{on right boundary where } x = 1, \\ hw + \frac{\partial w}{\partial y} & \text{on upper boundary where } y = 1, \end{cases}$$

which leads to the following solutions

$$w(x,y) = \begin{cases} C_1(y)\exp(h \cdot x) & \text{on left boundary where } x = 0, \\ C_2(x)\exp(h \cdot y) & \text{on lower boundary where } y = 0, \\ C_3(y)\exp(-h \cdot x) & \text{on right boundary where } x = 1, \\ C_4(x)\exp(-h \cdot y) & \text{on upper boundary where } y = 1. \end{cases}$$

Since $u(0) \geq u_0$ and $f \geq 0$ in Ω only non-negative functions C_1, C_2, C_3 and C_4 are logically possible. Then w is non-negative n the boundary $\partial\Omega$. □

The full heat conduction problem

The heat conduction problem under consideration is given by

$$\frac{\partial u(x,y,t)}{\partial t} - \alpha\Delta u(x,y,t) = f(x,y) \text{ and } u(x,y,0) = u_0(x,y) \text{ where } (x,y) \in \Omega = (0,1)^2 \text{ and } t \geq 0. \tag{35}$$

We supplement (35) with Robin boundary conditions:

$$h \cdot (u(x,y,t) - u_{\text{amb}}) + \frac{\partial u(x,y,t)}{\partial \nu_{\partial\Omega}} = 0 \text{ where } (x,y) \in \partial\Omega \text{ and } t \geq 0. \tag{36}$$

We will set our parameters as follows

Parameter	Value	Unit
α in (35)	0.1	m^2/s
$f(x,y)$ in (35)	as in (33)	-
S in $f(x,y)$	500	W/m^3
r in $f(x,y)$	0.1	m
h in (36)	0.5	$1/m$
u_{amb} in (36)	25.0	$°C$

Table 7: Problem parameters and their values.

Calculation of the Green function

The Green function as given by (28,29) is calculated as follows

- As integration method the summarized Milne rule is used. It is the closed Newton-Cotes formula of order 4. The rule is used separately for both spatial dimensions. We use 5001 sample points in each spatial direction.
- It turned out that using the first 414 eigenmodes leads to good results in the sense of the PDE residual.
- Since the heat source does not depend on the time t as well es the eigenmodes integration wrt. time and spatial integration allow for total decoupling. Time integration in (29) of the amplitude function can be done exactly and its integral allows the closed form $\int_0^t \exp\left(-\alpha(\mu_m^2 + \lambda_n^2)\tau\right) \mathrm{d}\tau = \frac{1-\exp\left(-\alpha(\mu_m^2+\lambda_n^2)t\right)}{\alpha(\mu_m^2+\lambda_n^2)}$.

Setup of the finite-difference solvers

PINN setup and first results

With the choice $u_0 = 25\,^\circ\mathrm{C}$, $a = 1/2$ and $b = 1$ in (34) we redo the simulation of case 1 where all other parameters are left unchanged.

Parameter	Model 1	Model 2	Model 3
Epochs	8,000	10,000	20,000
Samples	1,000	8,000	8,000
Step size	4,000	6,000	6,000
Relative error	7.86%	3.92 %	3.73 %

Table 8: Adjustment and Results.

First results

PDE Residuals (MAE) over Time

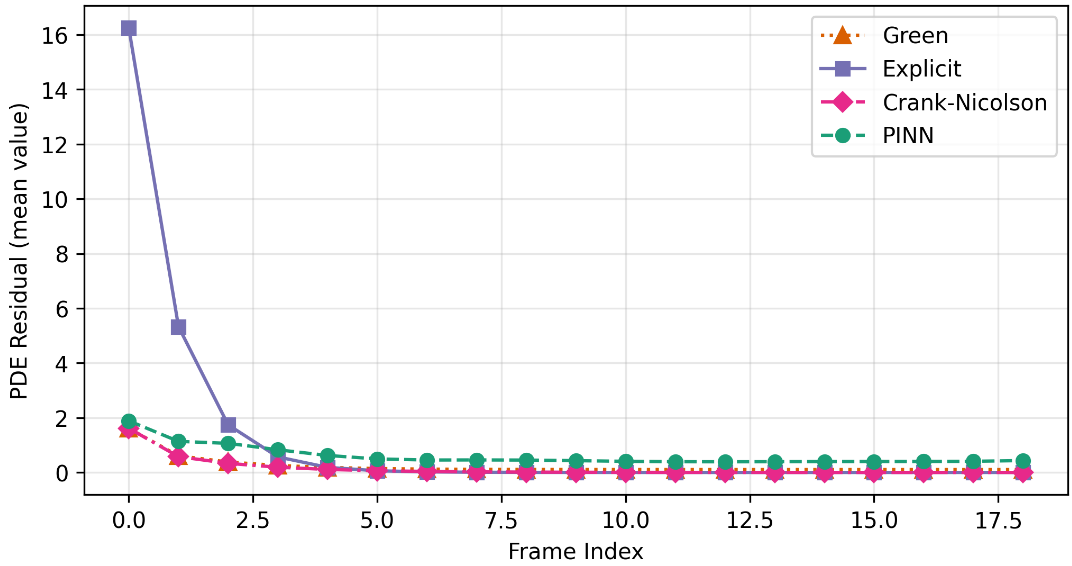


Figure 2: PDE Residuals (MAE) over time.

PDE Residuals (MAE) over Time

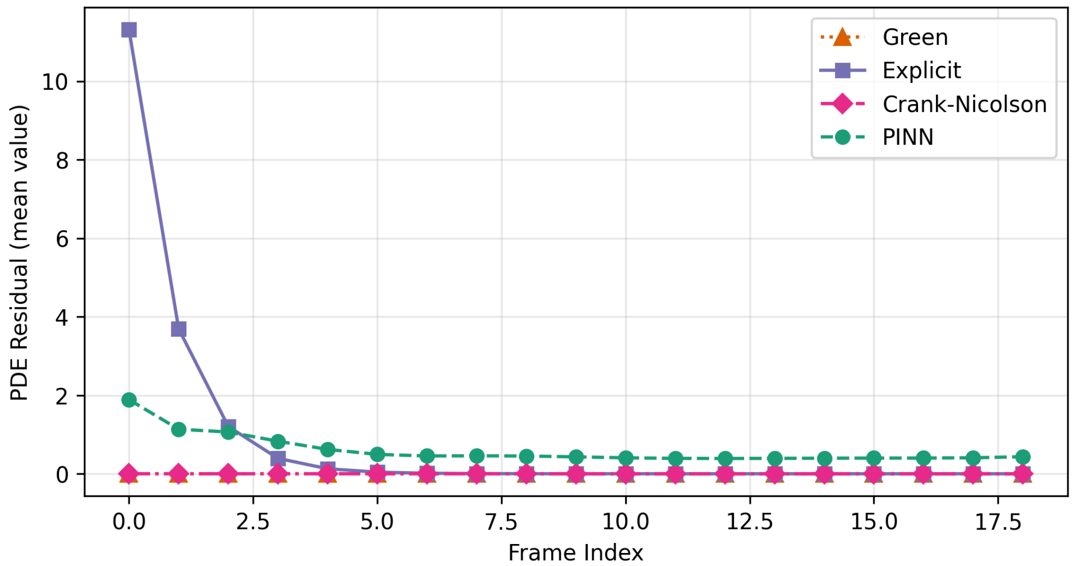
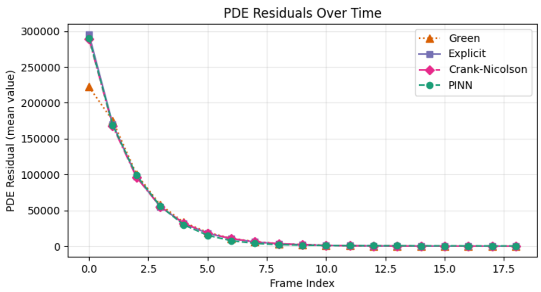


Figure 3: PDE Residuals (MAE) over time.

These solutions can be extended to t = 6 s. Then we get which promises convergence if t tends to infinity. The final frame at t = 6 s can be seen in the following heatmaps.

PDE	Boundary



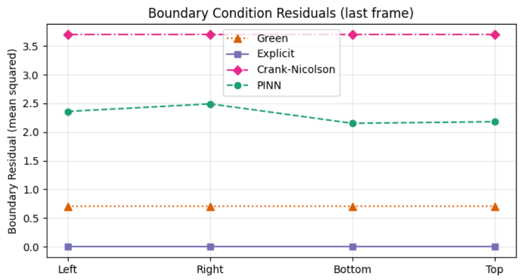


Table 9: Residuals.

Spatial Average of the calculated Temperatures

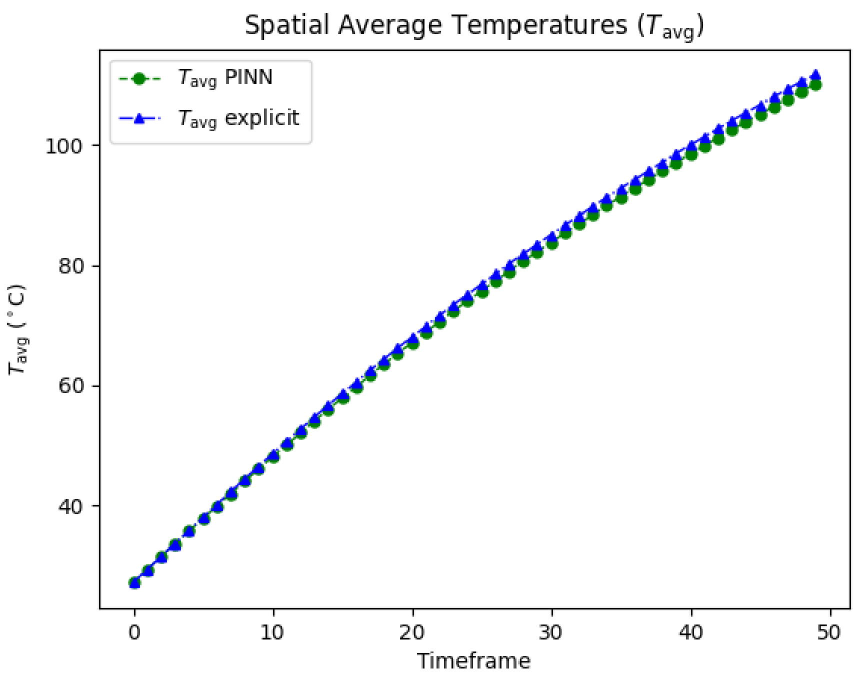


Table 10: Temperature averages for explicit and PINN solver.

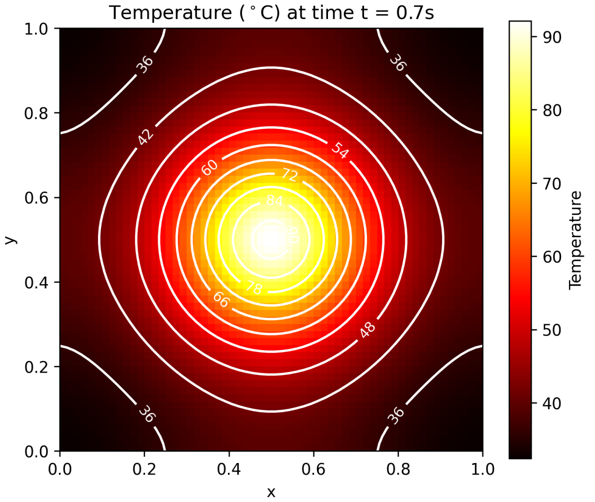
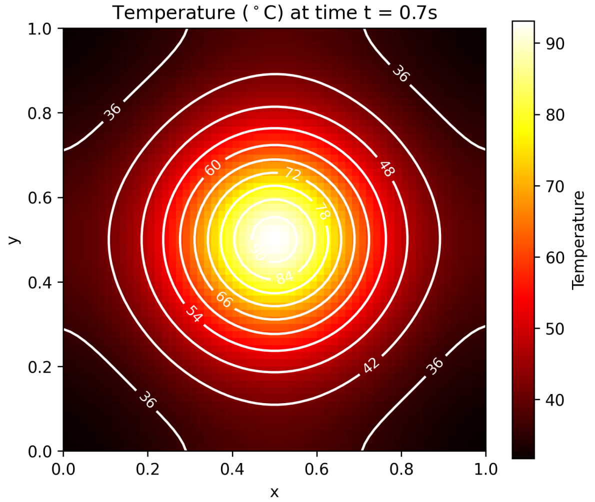
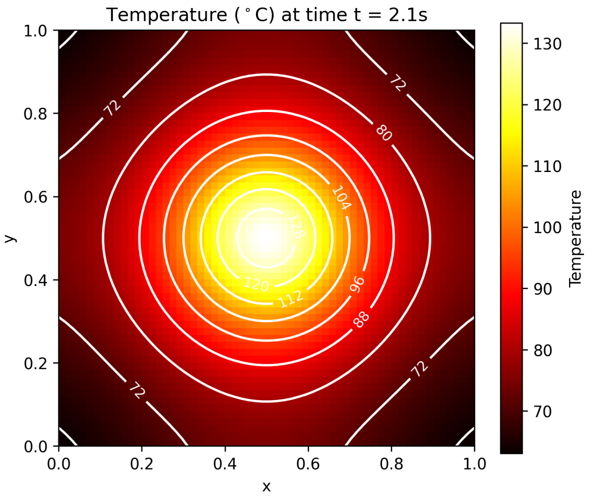
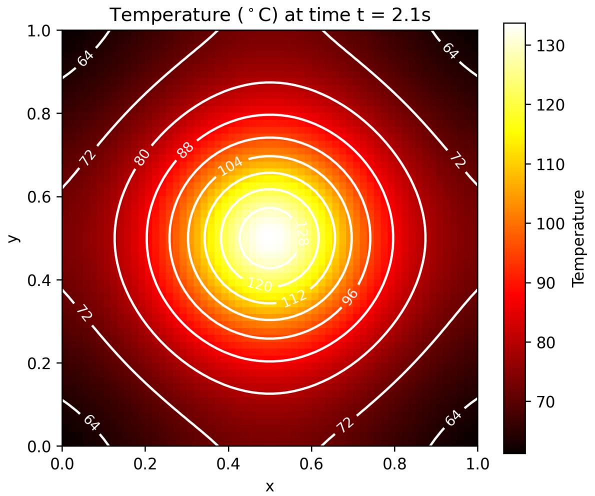
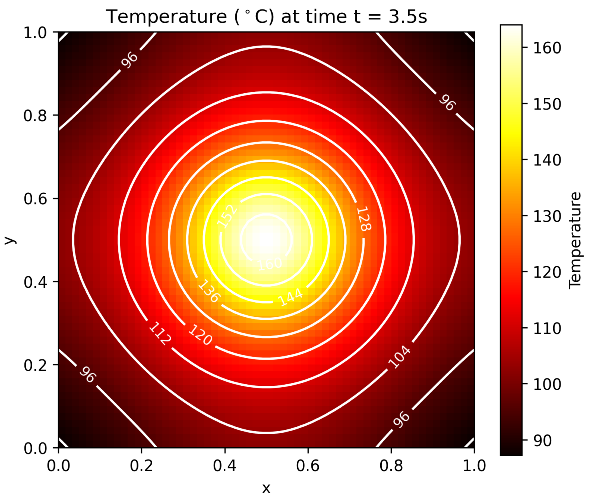
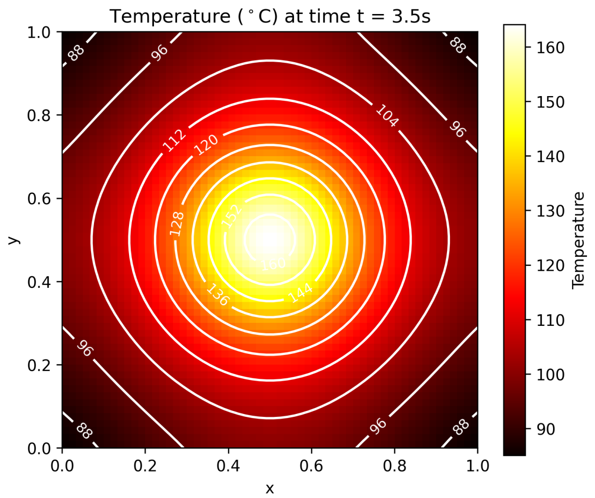
Explicit Solver	PINN
	
	
	

Table 11: Solution to different time frames.

PDE Residuals (MAE) over Time

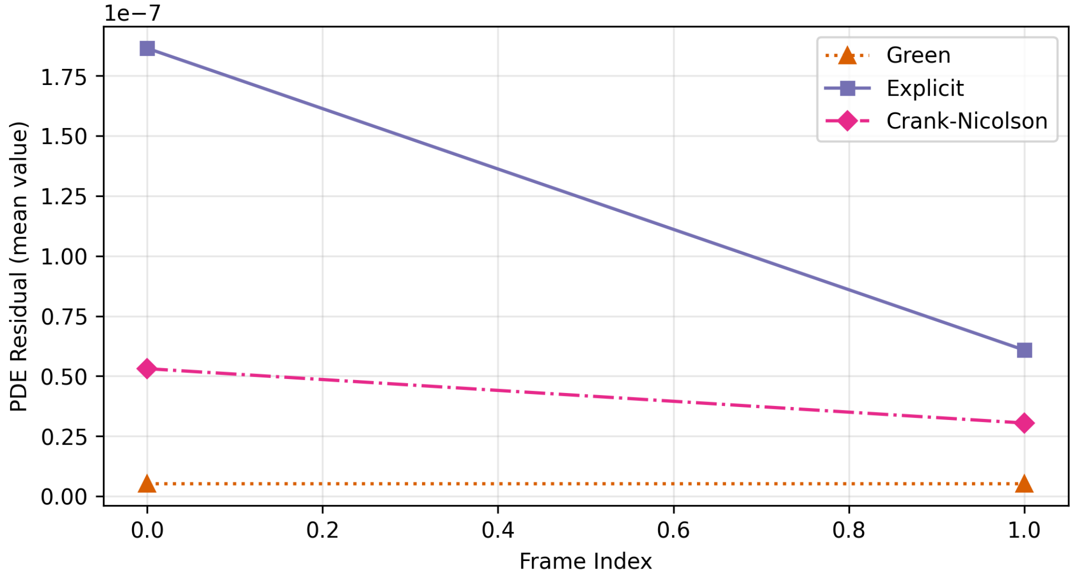


Figure 4: PDE Residuals (MAE) over time.

Boundary Residuals (MSE), last Frame

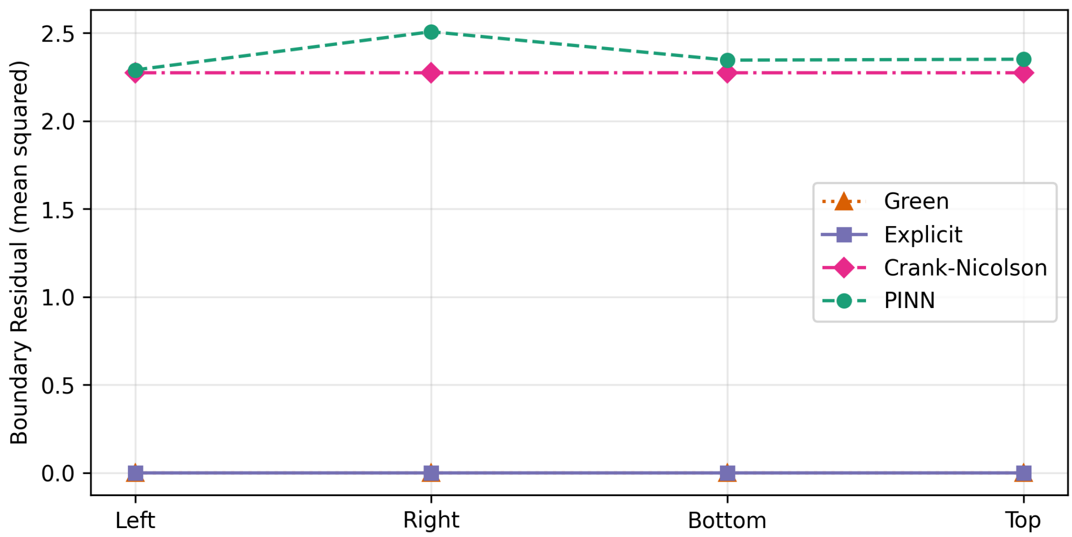


Figure 5: PDE Residuals (MAE) over time.

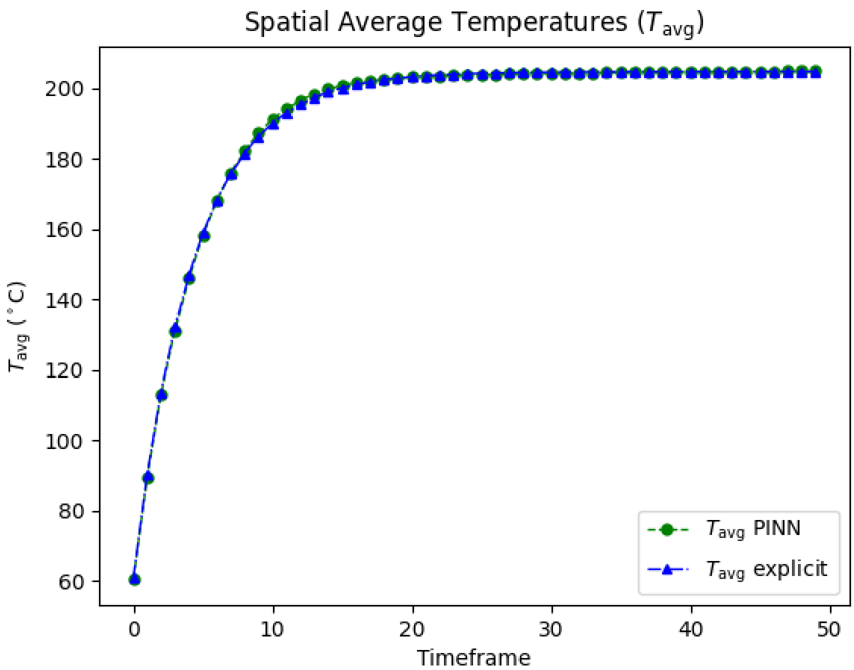
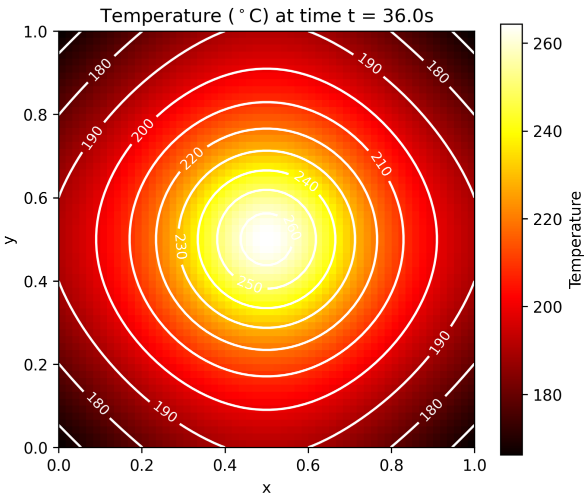
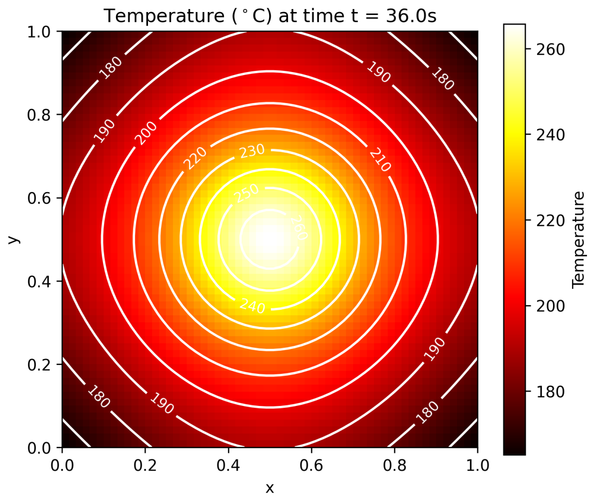
Spatial Average of the calculated Temperatures


Table 12: Temperature averages for explicit and PINN solver.

<table>
<tr><th>Explicit Solver</th><th>PINN</th></tr>
<tr><td>
</td><td>
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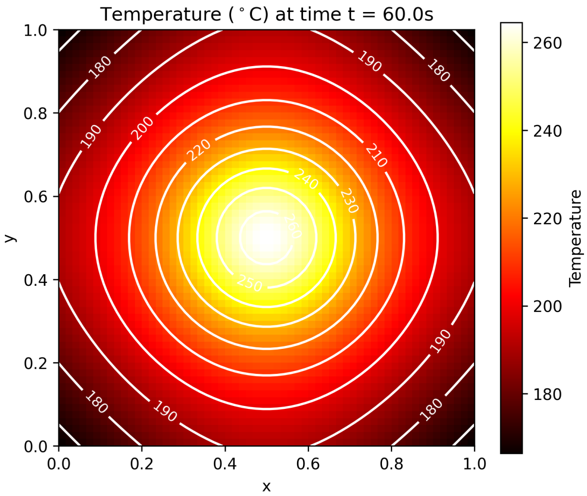
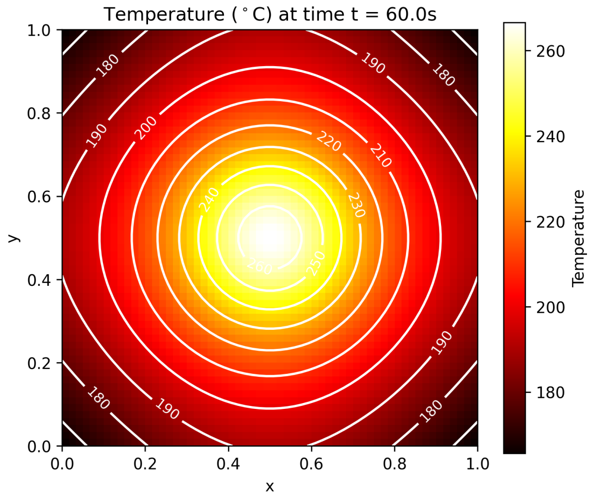
<tr><td>
</td><td>
</td></tr>
</table>

Table 13: Solution to different time frames.

References

[1] S. Aryal, *Simulation of 2-dimensional heat conduction with physics-informed neural networks.* Bachelor thesis, University of Applied Sciences Ravensburg-Weingarten, 2024

[2] S. Aryal, *Bachelor thesis: Pinns for 2d heat conduction.* Sourcecode of [1], Link (checked at 2025-09-10): https://github.com/Samman2571/Bachelor_thesis_PINNs_2D_Heat_Conduction

[3] I. N. Bronstein and K. A. Semendjajew, *Taschenbuch der Mathematik* 19., völlig neu bearb. Aufl., Gemeinschaftsausgabe Nauka, Moskau / BSB B. G. Teubner Verlagsgesellschaft, Leipzig, 1979; Lizenzausgabe, Verlag Harri Deutsch, Thun und Frankfurt am Main, 1980.

[4] A. Browder *Mathematical Analysis*, Corrected 3rd printing., Springer-Verlag Berlin Heidelberg New York, 2001.

[5] C. Eck, H. Garcke, and P. Knabner, *Mathematische Modellierung* 1st. ed., Springer-Verlag Berlin Heidelberg, 2008.

[6] O. Ernst, *Numerik partieller Differentialgleichungen, Part 2*, Lecture notes, University of Technology Chemnitz, 2015. Link (checked at 2025-09-10): https://www.tu-chemnitz.de/mathematik/numa/lehre/pde-2015/Folien/pdeTeil2.pdf

[7] H. Heuser, *Funktionalanalysis: Theorie und Anwendung*, 2. Aufl., Vieweg+Teubner Verlag, Wiesbaden, 1986.

[8] K.-H. Hoffmann and G. Witterstein, *Mathematische Modellierung: Grundprinzipien in Natur- und Ingenieurwissenschaften*, Mathematik Kompakt, 1st ed. (corrected reprint), Birkhäuser, Basel, 2014.

[9] R. López González *Neural Networks for Variational Problems in Engineering* PhD thesis, Technical University of Catalonia, 2008 [Online]. Available: https://digital.csic.es/bitstream/10261/310317/1/Tesis_Roberto_LÃşpez.pdf

[10] P. Meinhold and W. Wagner, *Partielle Differentialgleichungen*, BSB B. G. Teubner Verlagsgesellschaft, Leipzig, 1990.

[11] R. Meise and D. Vogt, *Einführung in die Funktionalanalysis* 1st ed., Friedr. Vieweg & Sohn Verlagsgesellschaft mbH, Braunschweig/Wiesbaden, 1992.

[12] G. D. Rey, and K. F. Wender *Neuronale Netze*, 2nd rev. and ext. ed., Verlag Hans Huber, Bern, 2011

[13] R. Rojas, *Theorie der neuronalen Netze: Eine systematische Einführung*, Springer-Lehrbuch, 1st ed. (corrected reprint 1996), Springer-Verlag, Berlin, 1993.

[14] W. Rudin *Principles of Mathematical Analysis*, 3rd ed., McGraw-Hill International Editions

[15] H. Stöcker, *Taschenbuch mathematischer Formeln und moderner Verfahren*, 3rd rev. and ext. ed., Harri Deutsch Verlag, Frankfurt am Main, 1995.

[16] W. Walter, *Gewöhnliche Differentialgleichungen*, 7. ed., Springer-Verlag Berlin Heidelberg, 2000.

[17] Code samples used in this work: https://github.com/Haasrobertgmxnet/HeatConduction